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Revision 2

Note: Just wanted to say, thanks for letting me submit this a little bit late, altium was giving me some really bad trouble the past couple weeks, but I had been keen on joining this even before the semester started (tysm) 🙏🙏 (finished on KiCad but will resolve before hopefully joining 🙏)

Also for the BOM CSV file its a bit strange and diff on KiCad which is why I made my own here as well

LM1117MP-ADJ/NOPB used. This was because the adjustable version made it so that we could choose 3.3V specifically. Used SOT 223 for the thermal and hand soldering balancing.

Deviated from the data sheets 100uf capacitor value in favor of something less, however with a better ESR for what we are looking for while still having a good margin to work with

Unsure about if a VIA needs to be used with every single capacitor ground, and how that works, along with if the math for T_J was done correctly since it was my first time learning it

Also for gerber and NC there is no 4.4 it is only 4.5 and 4.6 in kicad

BOM

Part	MPN	Digikey Part	Package / Case	Unit price	Datasheet	Lifecycle
Regulator	Already Given	Already Given	Already Given	Already Given	Already Given	Already Given
C1 / C2	TCTP1C10 6M8R	TCTP1C10 6M8R	0805	1.17	here	Active
C3	TLJR476M 006R3200	TLJR476M 006R3200	0805	1.02	here	Active
R1 / R3 (121)	RMCF080 5FT121R	RMCF080 5FT121R-ND	0805	0.11	here	Active
R2 (200)	RNCP080 5FTD200R	RNCP080 5FTD200R	0805	0.11	Link	Active
D1	XL-2012S URC	5962-XL-2 012SURC TR-ND	0805	0.01125 (for 3k)	Link	Active

J1, J2	M20-9990 246	M20-9990 246	2.54mm x 2	0.1	link	Active
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Section 4, Justification

Capacitors

- C1
 - 10uF (as mentioned on the datasheet)
 - Tantalum polarized capacitor
 - Voltage rating 16V
 - Net voltage ~5V
 - Derating ratio is approx 3. This is because of the ESR window for the tantalum and makes it so that it does not experience extreme heat, the reason for the 3x instead of 2x is just for stability and is because of any sort of noise or extra voltage that may occur or be placed on accident
- C3
 - Recommended from datasheet that if using C_{adj} that the uF value exceeds at least 22, which is why I chose 47 for the extra buffer. The voltage rating is also 6.3V which is good against the 3.3V that it experiences, giving it a 2 derating ratio
- C2
 - C_{adj} was said to be lower than 11uF in datasheet so we picked 10
 - Voltage across the node is around 2.05V (3.3V - 1.25 (reference voltage))
 - The reason for the overkill 16V is just so that we did not have to choose multiple types of capacitors since we could just get copies of the same one (would be easier for the long run and mean we would only have to buy 1 thing)

R1

- Default value on datasheet

R2

- Doing the math with the formula given $V_{out} = 1.25(1 + r_2 / r_1)$ and working backwards to find 3.3 Vout is 198ohms, however the closest is 200 ohms

R3

- Similar reason for value C2, since it means that the LED will work
- $(V_{out} - V_f) / I_{led} = R3$ which gave us 121ohms after assuming I_{led} as 10ma, since max is 20 and we want some buffer (sticking to the 2x trend as before)

D1

- Red + 0805 and V_f is 2.1V

J1 / J2

- Standard 2.54mm headers that would connect to most things, chosen specifically because they can handle the current and voltage ratings we are giving them, and also vertical.

5.1

- For R1 and R2 its 1% tolerance
- $V_{Ref} = 1.238 - 1.262$ from datasheet
- So what was done is choosing the combination of V_{ref} , R1 and R2 that makes $R2 / R1$ either the highest or lowest
- So min
 - $V_{REF} = 1.238$, $R2 = 198$, $R1 = 122.21$ using equation from before is 3.24V
- So max
 - $V_{REF} = 1.262$, $R2 = 202$, $R1 = 119.79$ using equation from before is 3.39V
- So absolute worst case is 3.24 - 3.39V when accounting for resistor tolerance and ref voltage
- If ADJ is floating then the entire thing would not be controlled and not be regulated, since the equation we have been using will become invalid

5.2 Thermal

- $P = (V_{in} - v_{out}) \times \text{load}$
- At 500ma and 5v / 3.24v respectively we get 0.88W
- Efficiency = $v_{out} / v_{in} = 66\%$ using a nominal v_{out}
- Theta JA = drops as copper increases, we chose this because when finding the temperatures at 45 and 25 degrees we got theta of 133.24 and 142.05 using
 - $T_J = T_A + P \times \text{Theta}_{JA}$ where T_J is never allowed over 150 and T_A is given to us and $P + \text{Theta}_{JA}$ are given to us via chart and calculations

5.3

- Base minimum is 10uf and ESR should be between 0.3 and 22 ohm
- Since I am using C_{adj} the min goes up to 22 uf
- Cannot use ceramics since ESR too low and not polarized and found something with 3.2 ohm which is in the range

7.4

- The nets that carry the load current are 5V 3V3 and the GND carry higher AMPS
- The LED branch has much less current that we calculated earlier

- Using the table we get around 10-12mil as the minimum, but that seems to be the BARE minimum
- The actual choice was more similar to the minimum that we were supposed to give (0.25mm) and i decided to increase it to 0.7 for the sake of low risk and we are not worried about route congestion and meet the requirements for board size

7.5

- Based off previous calculations and working with 0.066inch^2 for the thermal, I made a block that was a bit bigger, around 74mm^2 which is around 30mm^2 greater then what was required, just to ease out the margins a bit since it seemed a bit tighter